What would you like to know about this coin?



Thirty observed tosses. What is still unknown?

What would you like to know about this coin?

H H T H T H T H H H T T H T H

H T H T T H H H T T H T H T H

r is the unknown chance of heads.

> Do we want one guess, or uncertainty about the guess?

The data are fixed; the unknown is not a data point

Observed $\mathcal{D}$

$$17 \text{ heads}, \quad 13 \text{ tails}$$

Unknown r

$$0 < r < 1$$

> Which object is observed? Which object will have a posterior?

The data are fixed; the unknown is not a data point

Observed $\mathcal{D}$

$$17 \text{ heads}, \quad 13 \text{ tails}$$

Unknown r

$$0 < r < 1$$

Inference describes plausible values of r.

> Does inferring r change any of the thirty tosses?

A prior is a distribution over the unknown

$$r \sim \mathrm{Beta}(5, 3)$$

> What does the Beta prior describe?

A prior is a distribution over the unknown

$$r \sim \mathrm{Beta}(5, 3)$$

$$p(r \mid \mathcal{D}) \;\propto\; p(\mathcal{D} \mid r)\, p(r)$$

> What two ingredients turn it into a posterior shape?

What normalises the posterior?

$$p(r \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid r)\, p(r)}{p(\mathcal{D})}$$

> What is the denominator doing?

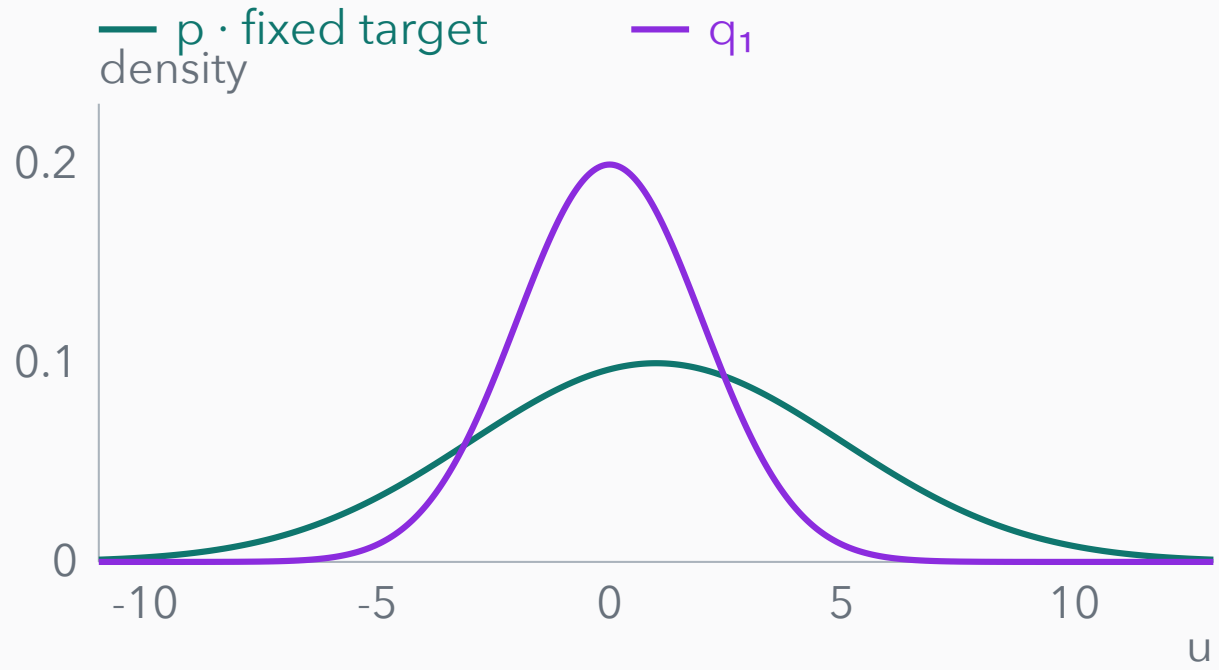
What normalises the posterior?

$$p(r \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid r)\, p(r)}{p(\mathcal{D})}$$

This coin is solvable. We will use its answer to check a more general method.

> Why practise an approximate method on a solvable problem?

Which Gaussian is closer to the target?



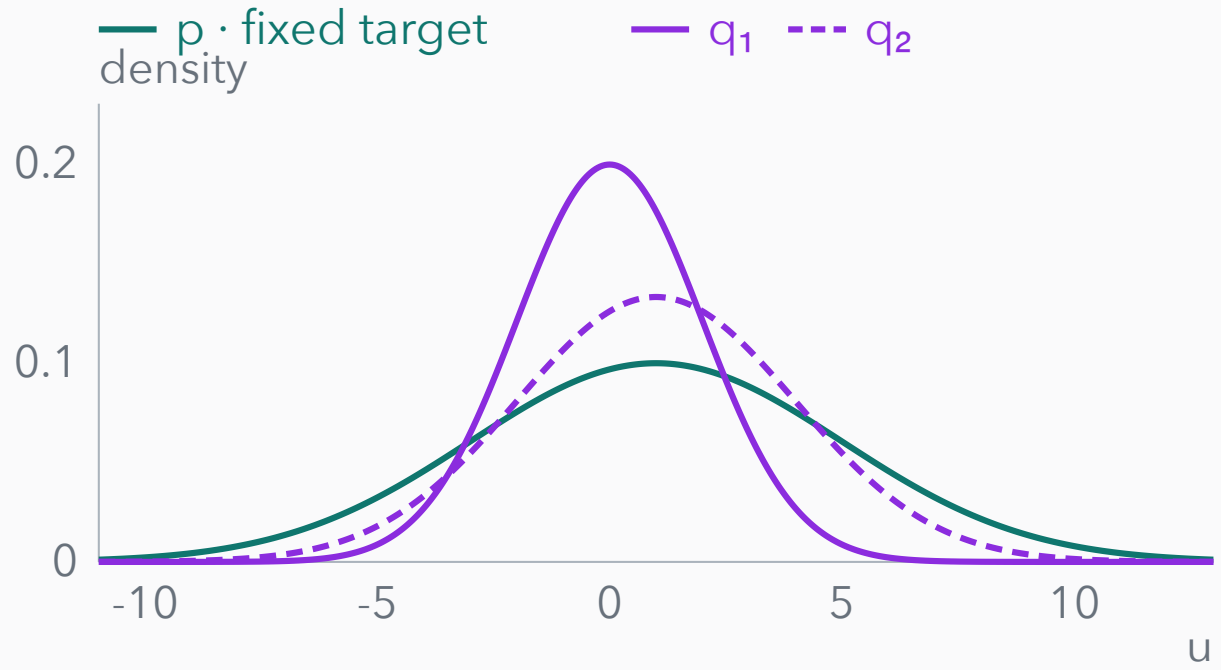
$$p = \mathcal{N}(1, 4^2)$$

$$q_1 = \mathcal{N}(0, 2^2)$$

Normal(loc=μ, scale=σ). The mathematical second argument is variance.

Describe the centre and width of q_1.

Which Gaussian is closer to the target?



$$p = \mathcal{N}(1, 4^2)$$

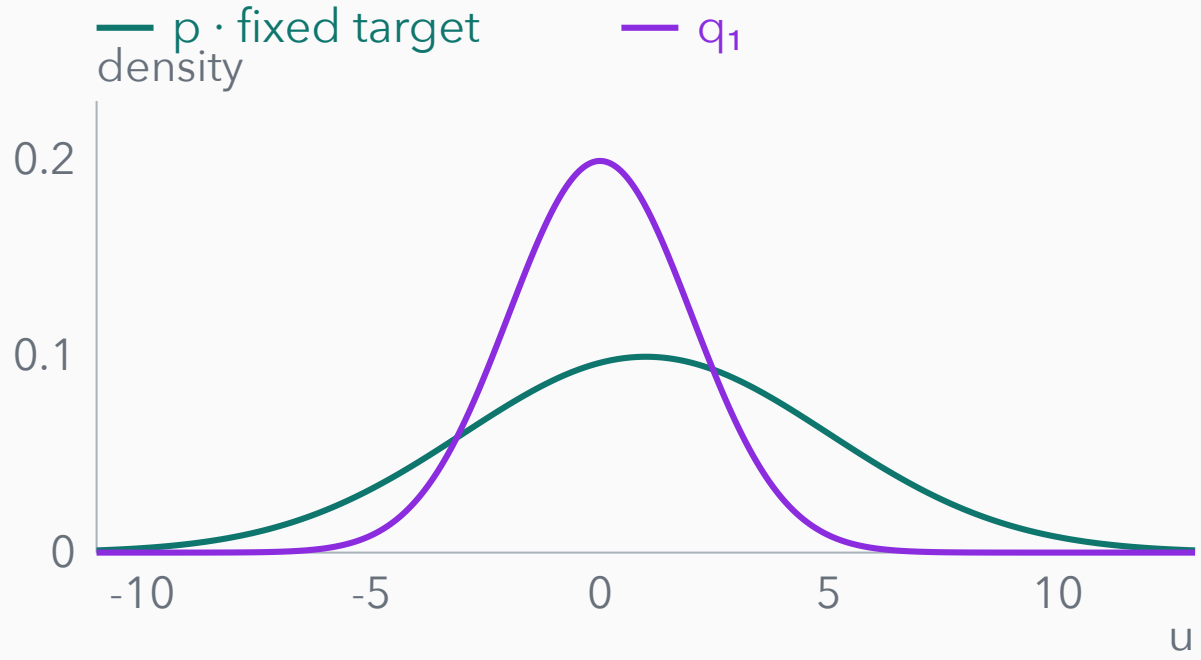
$$q_1 = \mathcal{N}(0, 2^2)$$

$$q_2 = \mathcal{N}(1, 3^2)$$

Normal(loc=μ, scale=σ). The mathematical second argument is variance.

Vote for q_1 or q_2 before calculating.

One density comparison is not the whole story

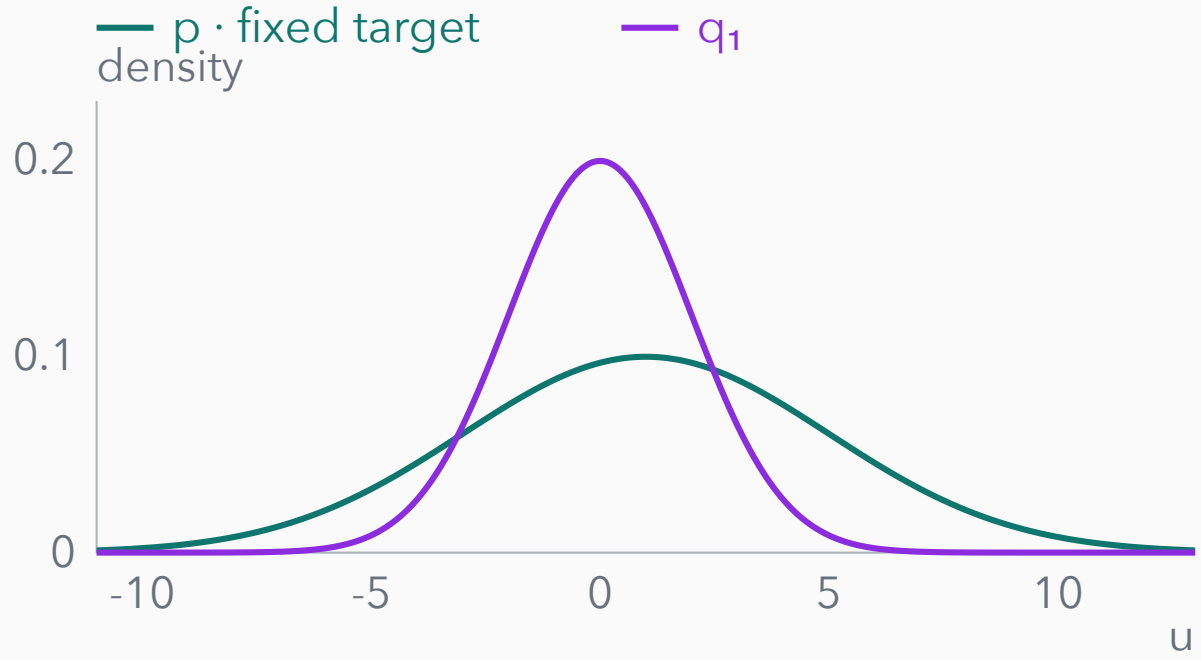


$$u = 0$$

$$\log q_1(0) - \log p(0)$$

At u=0, which density is larger?

One density comparison is not the whole story



$$u = 0$$

$$\log q_1(0) - \log p(0)$$

This compares one location. What about the rest of the curve?

> Could this one location settle which whole distribution is closer?

Average the log ratio under q

$$\mathrm{KL}(q\|p) = \mathbb{E}_{u\sim q}[\log q(u) - \log p(u)]$$

> Where does this expectation get its values of u?

Average the log ratio under q

$$\mathrm{KL}(q\|p) = \mathbb{E}_{u\sim q}[\log q(u) - \log p(u)]$$

The first argument supplies the weighting distribution.

> What does the order q, then p tell us?

Reversing KL changes the question

$$\mathrm{KL}(q_1 \| p)$$

$$\mathrm{KL}(p \| q_1)$$

> Are these two averages weighted in the same way?

Reversing KL changes the question

$$\mathrm{KL}(q_1 \| p)$$

$$\approx 0.3494$$

$$\mathrm{KL}(p \| q_1)$$

$$\approx 0.9319$$

> What does the numerical difference rule out?

Nonnegative KL does not make every term positive

$$\log q(u) - \log p(u) \quad \text{can be negative}$$

> Can p(u) exceed q(u) at a particular u?

Nonnegative KL does not make every term positive

$$\log q(u) - \log p(u) \quad \text{can be negative}$$

$$\mathrm{KL}(q\|p) \geq 0, \qquad \mathrm{KL}(p\|p) = 0$$

Which nonnegative statement applies to the whole average?

For two Gaussians, evaluate the expectation exactly

$$q = \mathcal{N}(\mu, \sigma^2), \qquad p = \mathcal{N}(m, s^2)$$

> Which parameters belong to q, and which stay fixed in p?

For two Gaussians, evaluate the expectation exactly

$$q = \mathcal{N}(\mu, \sigma^2), \qquad p = \mathcal{N}(m, s^2)$$

$$\mathrm{KL}(q\|p) = \log \frac{s}{\sigma} + \frac{\sigma^2 + (\mu - m)^2}{2s^2} - \frac{1}{2}$$

> Where do width and centre mismatch enter?

Substitute first; then compare the score

$$q_1: \quad \log 2 + \frac{4+1}{32} - \frac{1}{2} \qquad\qquad q_2: \quad \log \frac{4}{3} + \frac{9+0}{32} - \frac{1}{2}$$

> Which candidate has no squared centre-shift term?

Substitute first; then compare the score

$$q_1: \quad \log 2 + \frac{4+1}{32} - \frac{1}{2}$$

$$\mathrm{KL}(q_1 \| p) \approx 0.3494$$

$$q_2: \quad \log \frac{4}{3} + \frac{9+0}{32} - \frac{1}{2}$$

$$\mathrm{KL}(q_2 \| p) \approx 0.0689$$

> Does the score support the earlier prediction?

A comparison can become a search

Keep the target fixed. Change q’s parameters.

> What should the optimiser change?

A comparison can become a search

Keep the target fixed. Change q’s parameters.

$$\psi^* = \arg\min_{\psi} \mathrm{KL}(q_\psi \| p)$$

> What does this argmin select?

A moment identity explains the squared shift

$$u - m = (u - \mu) + (\mu - m)$$

When expanding the square, which cross-term should vanish?

A moment identity explains the squared shift

$$u - m = (u - \mu) + (\mu - m)$$

$$\mathbb{E}_q[(u - m)^2] = \mathbb{E}_q[(u - \mu)^2] + (\mu - m)^2$$

Why does the remaining random term become σ²?

Move the centre while keeping the width fixed

$$\mu = 0, \quad \sigma = 2, \qquad p = \mathcal{N}(1, 4^2)$$

> If the width cannot change, which way should μ move?

Move the centre while keeping the width fixed

$$\mu = 0, \quad \sigma = 2, \qquad p = \mathcal{N}(1, 4^2)$$

$$L(\mu) = \underbrace{\log 2 + \frac{4}{32} - \frac{1}{2}}_{\text{constant}} + \frac{(\mu - 1)^2}{32}$$

Which term changes when μ changes?

A negative derivative produces a rightward descent step

$$\frac{\partial L}{\partial \mu} = \frac{2(\mu - 1)}{32}$$

> Differentiate the squared shift.

A negative derivative produces a rightward descent step

$$\frac{\partial L}{\partial \mu} = \frac{2(\mu - 1)}{32}$$

$$= \frac{\mu - 1}{16}$$

What simplifies in the derivative?

A negative derivative produces a rightward descent step

$$\frac{\partial L}{\partial \mu} = \frac{2(\mu - 1)}{32}$$

$$= \frac{\mu - 1}{16}$$

$$\left.\frac{\partial L}{\partial \mu}\right|_{\mu=0} = -0.0625$$

> Does a negative derivative mean move left under gradient descent?

One literal update

$$\mu^{+} = \mu - \eta \frac{\partial L}{\partial \mu}$$

> What does gradient descent subtract?

One literal update

$$\mu^{+} = \mu - \eta \frac{\partial L}{\partial \mu}$$

$$\eta = 4, \qquad \mu^{+} = 0 - 4(-0.0625)$$

> Predict the new mean.

One literal update

$$\mu^{+} = \mu - \eta \frac{\partial L}{\partial \mu}$$

$$\eta = 4, \qquad \mu^{+} = 0 - 4(-0.0625)$$

$$\mu^{+} = 0.25$$

> Which parameter has not changed?

The new curve earns a new loss evaluation



$$\mu : 0 \longrightarrow 0.25$$

σ=2 throughout. Same target and axes.

What changed in the curve?

The new curve earns a new loss evaluation



$$\mu : 0 \longrightarrow 0.25$$

KL before	KL after
0.3494	0.3357

σ=2 throughout. Same target and axes.

Did the computed objective improve?

Perfect mean fitting still leaves a gap

$$\mu = 1, \quad \sigma = 2 \neq 4$$

> What is the best mean if σ must stay 2?

Perfect mean fitting still leaves a gap

$$\mu = 1, \quad \sigma = 2 \neq 4$$

$$\min_{\mu} \mathrm{KL}(q_\mu \| p) \approx 0.3181 > 0$$

> Why is the minimum KL still positive?

A standard deviation must remain positive

$$\rho \in \mathbb{R}, \qquad \sigma = \exp(\rho) > 0$$

> Why not update an unconstrained σ directly?

A standard deviation must remain positive

$$\rho \in \mathbb{R}, \qquad \sigma = \exp(\rho) > 0$$

```
mu = Parameter(tensor(0.))
rho = Parameter(tensor(log(2.)))
q = Normal(mu, rho.exp())
```

What does ρ represent here?

Rebuild, score, differentiate, update

```
q = Normal(mu, rho.exp())
loss = kl_divergence(q, p)
optimiser.zero_grad()
loss.backward()
optimiser.step()
```

Why build q again on the next iteration?

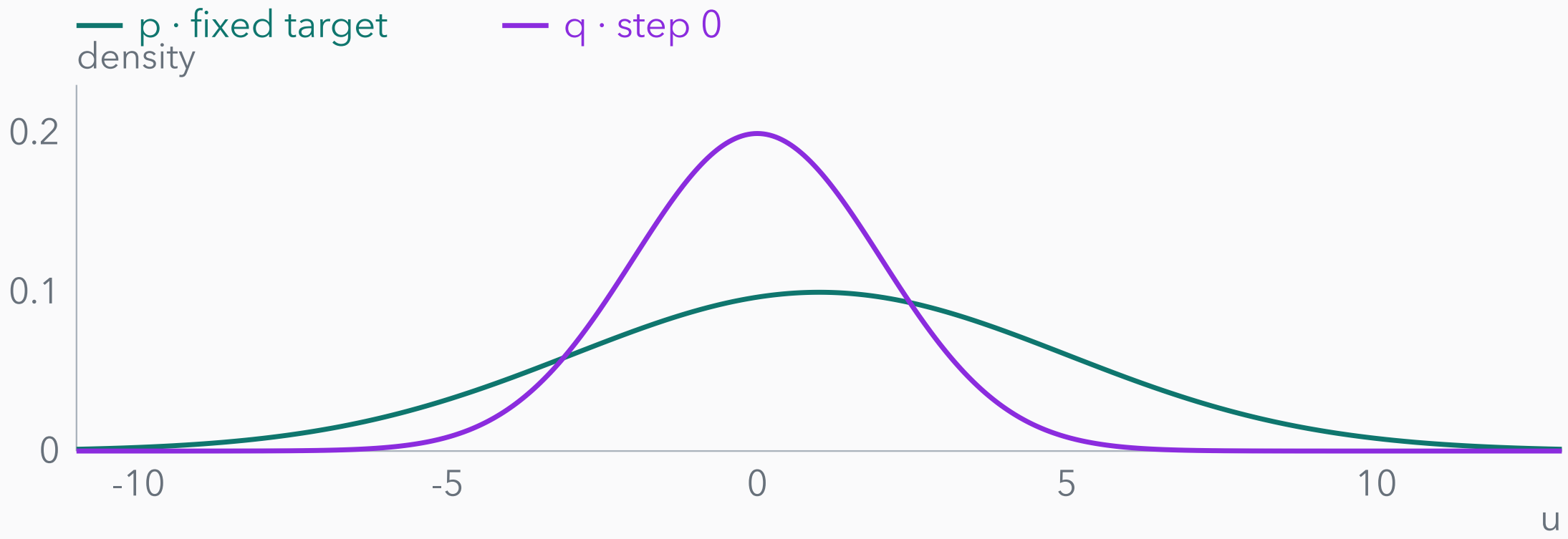
Rebuild, score, differentiate, update

```
q = Normal(mu, rho.exp())
loss = kl_divergence(q, p)
optimiser.zero_grad()
loss.backward()
optimiser.step()
```

Repeat with the updated parameters.

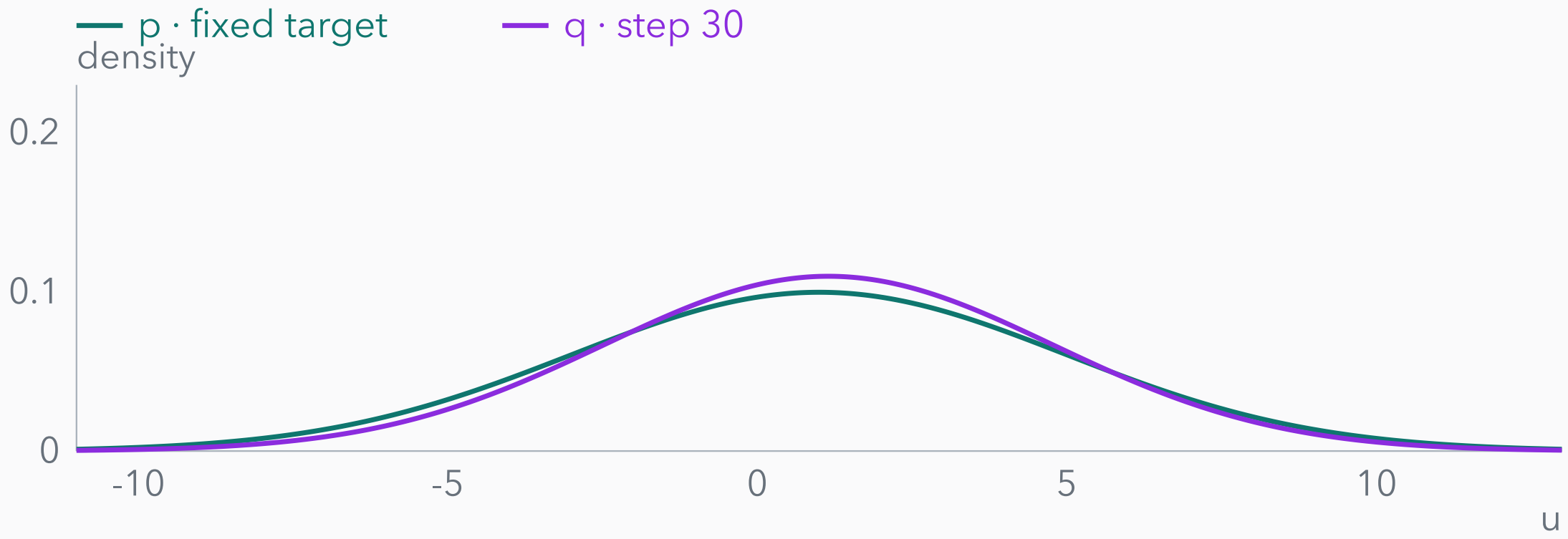
> Why clear gradients before backward?

Letting both parameters move removes the restriction



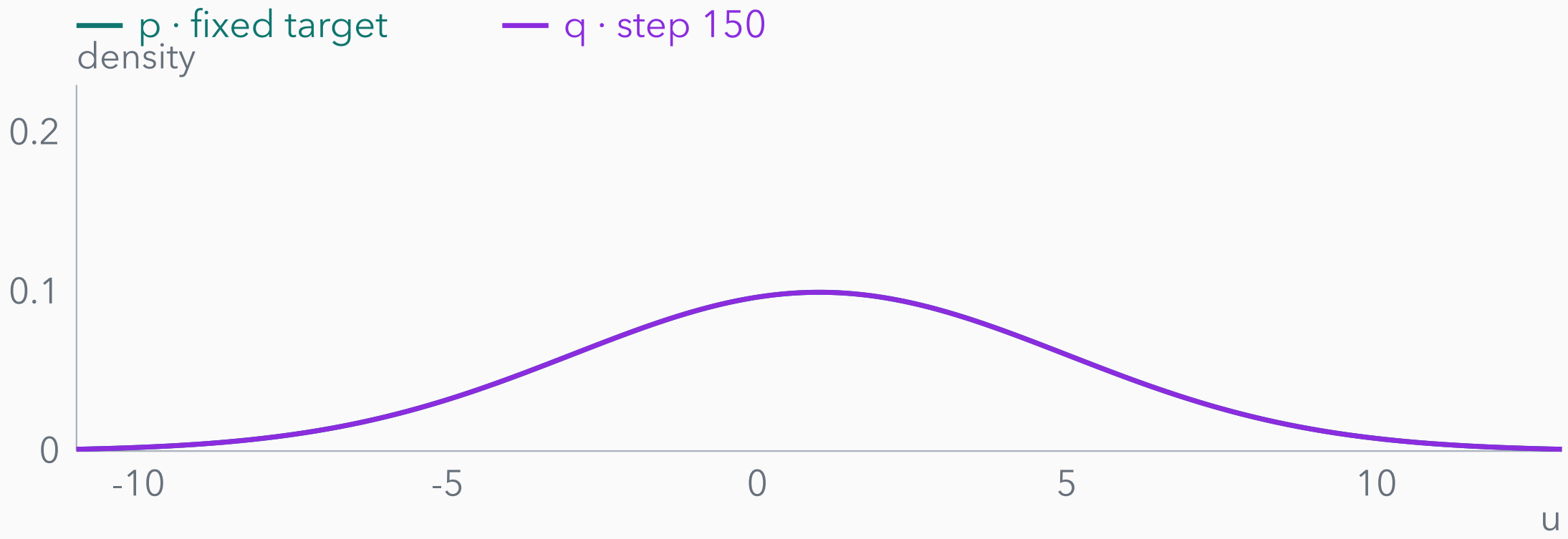
What can this larger family now match?

Letting both parameters move removes the restriction



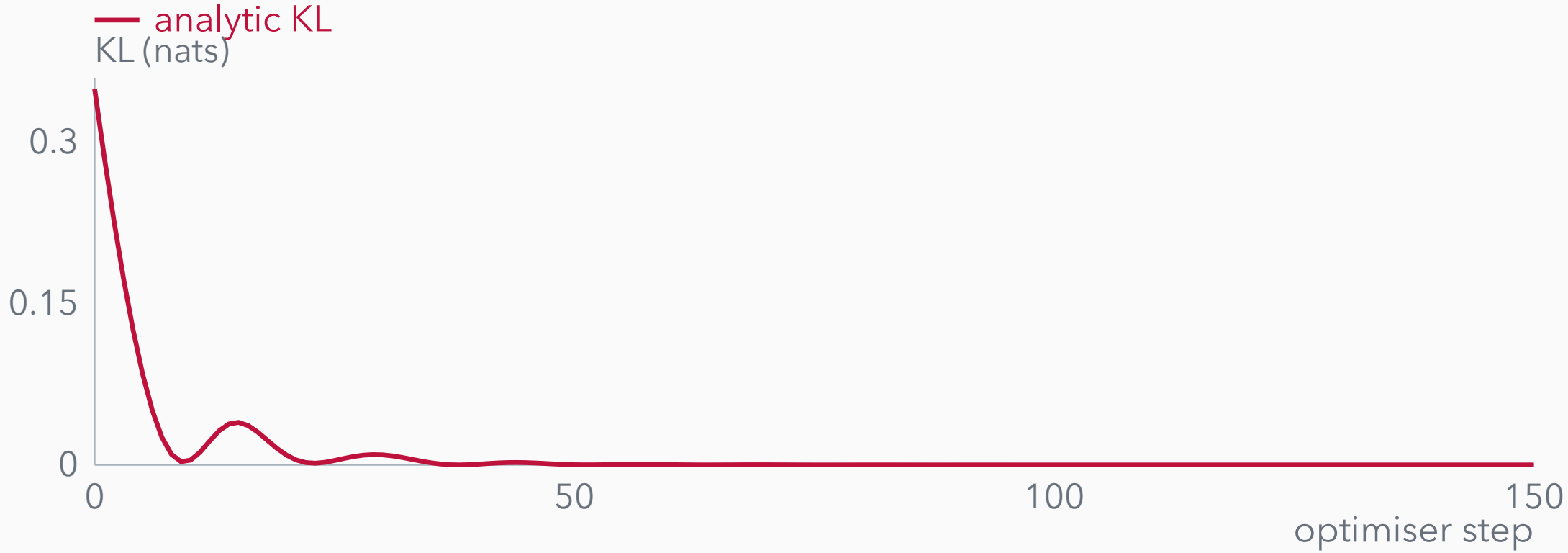
At step 30, what is still visibly different?

Letting both parameters move removes the restriction



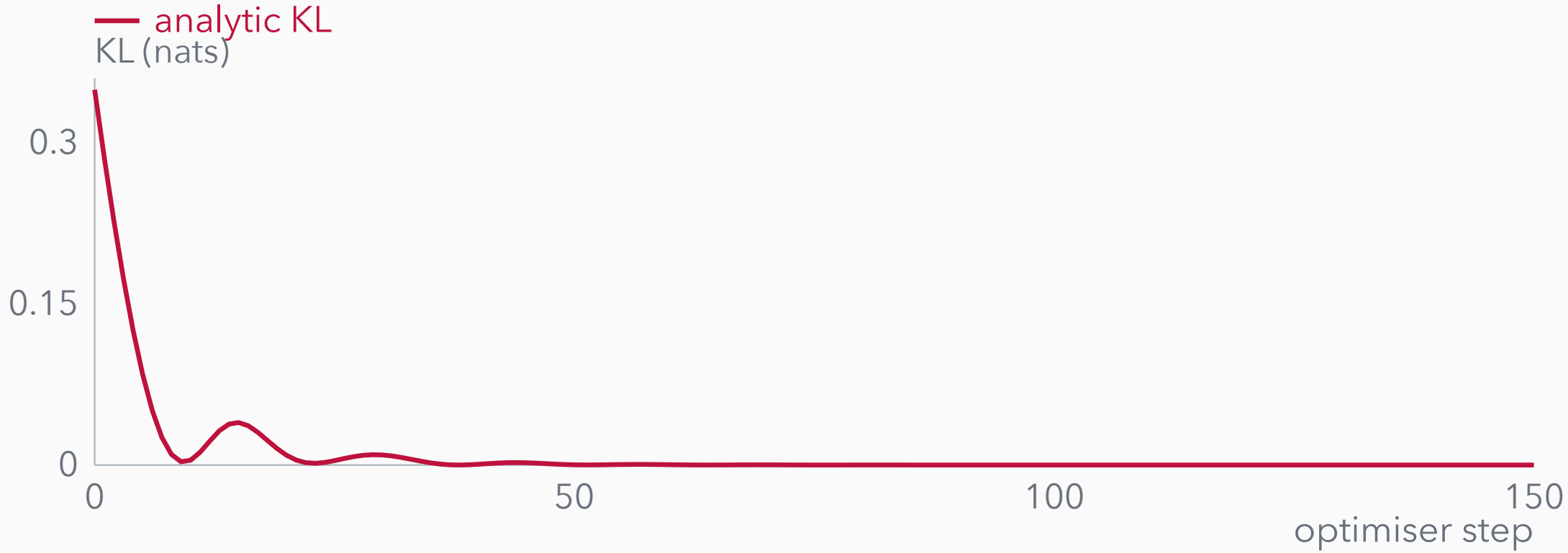
How close is the final measured fit?

A loss trace answers a specific question



What is being measured on the vertical axis?

A loss trace answers a specific question



This trace evaluates the exact Normal formula. It is not a Monte Carlo estimate.

> Would the same smooth trace be guaranteed for a sampled objective?

Replace the target; keep the definition

$$p(u) = \tfrac{1}{2}\mathcal{N}(u; -0.2, 0.4^2) + \tfrac{1}{2}\mathcal{N}(u; 1, 0.5^2)$$

> What makes this a normalised density?

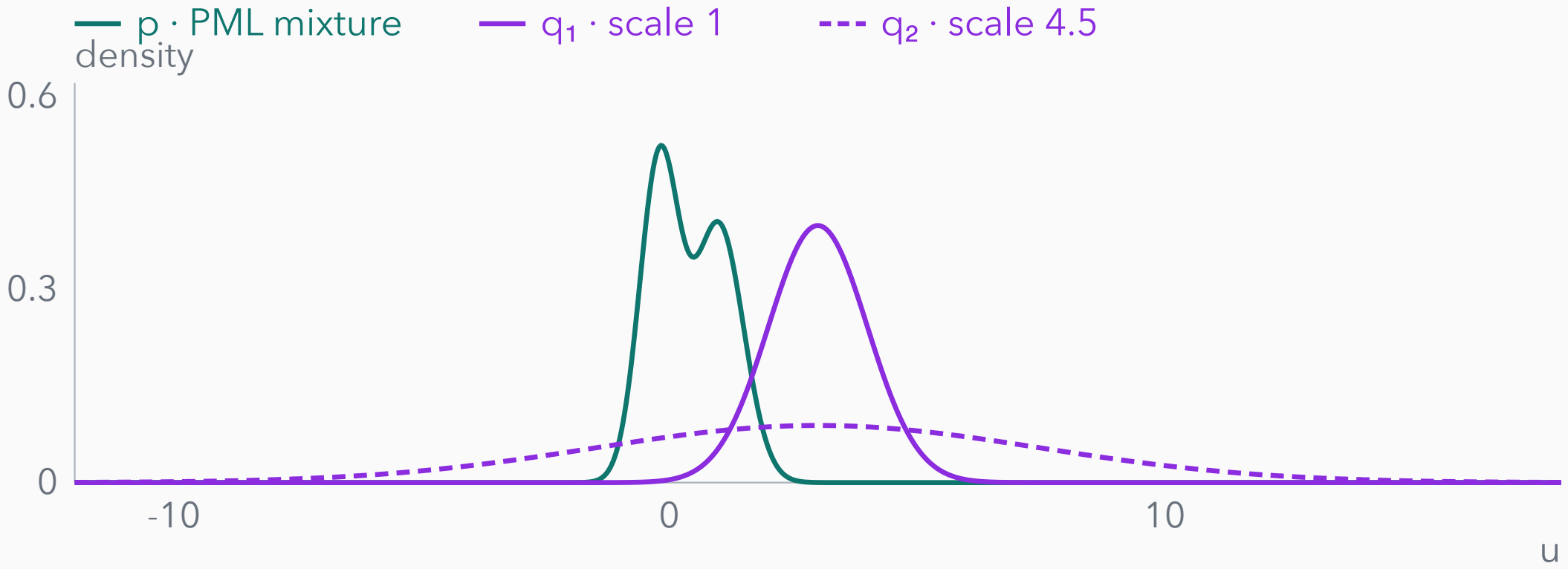
Replace the target; keep the definition

$$p(u) = \tfrac{1}{2}\mathcal{N}(u; -0.2, 0.4^2) + \tfrac{1}{2}\mathcal{N}(u; 1, 0.5^2)$$

We can evaluate this density at a proposed u.

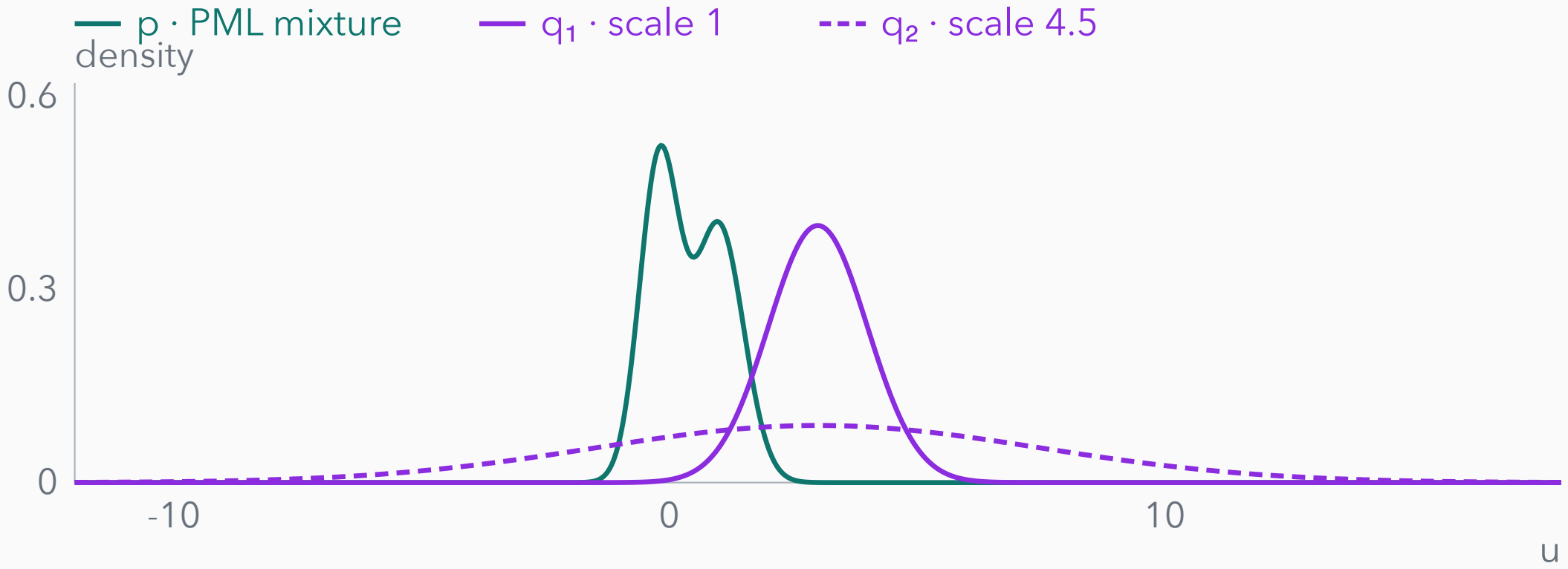
> Does evaluating p(u) require a closed-form KL?

Which candidate wastes more mass far from the mixture?



Where does p assign most of its density?

Which candidate wastes more mass far from the mixture?



Both candidates have mean 3. Their standard deviations are 1 and 4.5.

Which candidate draws more often from far-away regions?

A missing library formula is not an undefined quantity

```
kl_divergence(q, mixture)
```

What would this call need to know?

A missing library formula is not an undefined quantity

```
kl_divergence(q, mixture)
```

```
NotImplementedError:
KL not implemented for this distribution pair
```

What does this error establish, and what does it not establish?

Draw from q, score both densities, average

$$u_1, \ldots, u_S \sim q$$

> Which distribution supplies the draws?

Draw from q, score both densities, average

$$u_1, \ldots, u_S \sim q$$

$$\ell_i = \log q(u_i) - \log p(u_i)$$

What can we compute at each draw?

Draw from q, score both densities, average

$$u_1, \ldots, u_S \sim q$$

$$\ell_i = \log q(u_i) - \log p(u_i)$$

$$\widehat{\mathrm{KL}}_S = \frac{1}{S} \sum_{i=1}^{S} \ell_i$$

Which dimension is averaged?

Validate the estimator before using the unfamiliar target

$$q = \mathcal{N}(1, 3^2), \quad p = \mathcal{N}(1, 4^2)$$

```
u = q.rsample((S,))
log_q = q.log_prob(u)
log_p = p.log_prob(u)
estimate = (log_q-log_p).mean()
```

S=10,000 · seed 1000

What result can we check without Monte Carlo?

Validate the estimator before using the unfamiliar target

```
u = q.rsample((S,))
log_q = q.log_prob(u)
log_p = p.log_prob(u)
estimate = (log_q-log_p).mean()
```

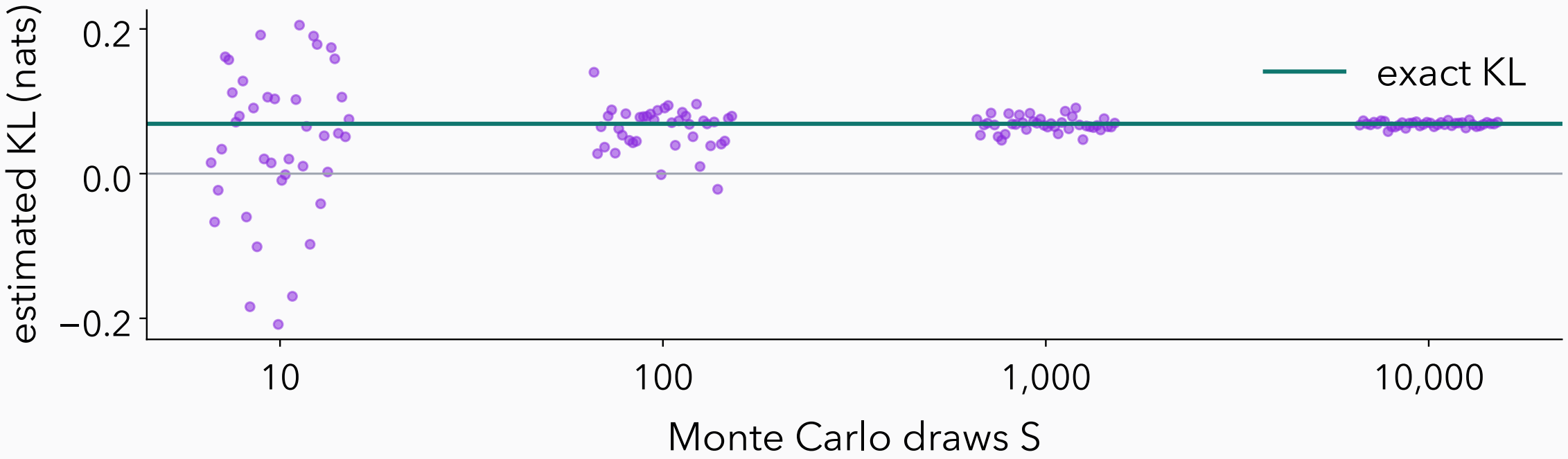
$$q = \mathcal{N}(1, 3^2), \quad p = \mathcal{N}(1, 4^2)$$

Exact KL	One MC estimate
0.0689	0.0671

S=10,000 · seed 1000

Should a finite estimate equal the exact answer digit for digit?

More draws reduce variability, not every individual error

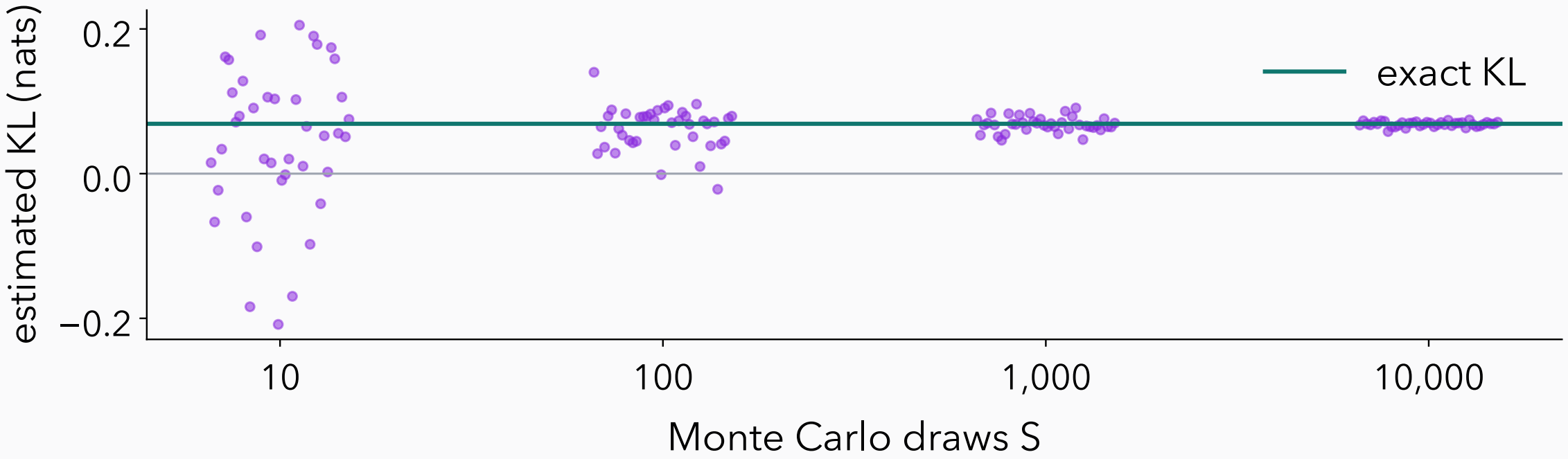


Same p and q · 40 seeds per sample count · exact KL is the horizontal reference

> What changed between these four groups?

Measured estimates · seeds1000–1039 · S=10,100,1000,10000 · float64

More draws reduce variability, not every individual error



Same p and q · 40 seeds per sample count · exact KL is the horizontal reference

A new seed changes the estimate. It does not change the exact KL.

> Must every larger-S estimate be closer than every smaller-S estimate?

Measured estimates · seeds1000–1039 · S=10,100,1000,10000 · float64

Five draws can give a negative KL estimate

u_i	$\log q(u_i) - \log p(u_i)$
5.623	−0.232
0.120	0.269
−5.536	−0.751
2.705	0.217
−2.254	0.030

> Some rows are negative. Could their average be negative?

Five draws can give a negative KL estimate

u_i	$\log q(u_i) - \log p(u_i)$
5.623	−0.232
0.120	0.269
−5.536	−0.751
2.705	0.217
−2.254	0.030

$$\widehat{\mathrm{KL}}_5 \approx -0.0933, \qquad \mathrm{KL}(q\|p) \approx 0.0689$$

> Does this contradict nonnegative exact KL?

The estimated score separates these candidates clearly

Candidate	MC estimate ± one SE	Numerical integral
q1: σ=1	9.545 ± 0.026	9.489
q2: σ=4.5	45.023 ± 0.212	45.152

> How large is the difference relative to the sampling uncertainty?

100,000 q draws · seeds402/403 · separate wide-domain numerical integration

The estimated score separates these candidates clearly

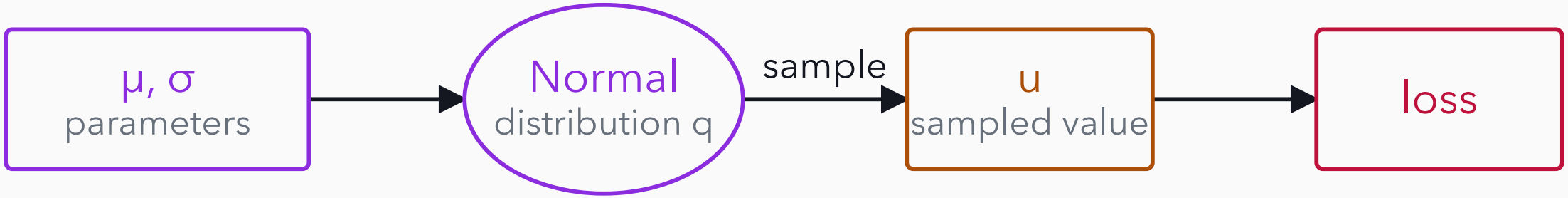
Candidate	MC estimate ± one SE	Numerical integral
q1: σ=1	9.545 ± 0.026	9.489
q2: σ=4.5	45.023 ± 0.212	45.152

q_1 has the smaller directional KL. The estimate is not an analytic formula.

> What role does the numerical integral play?

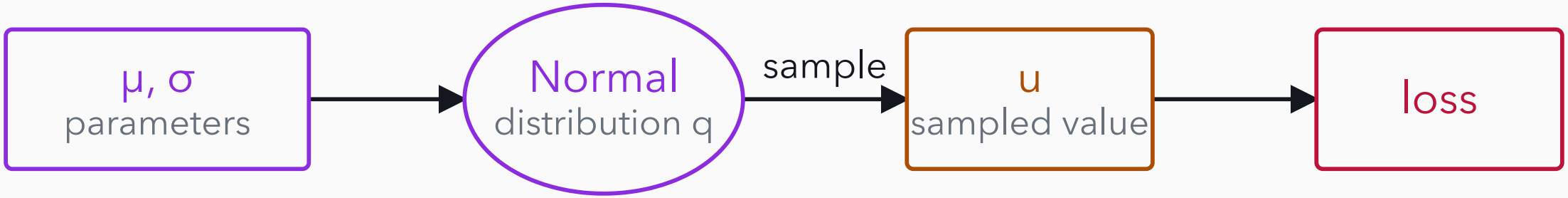
100,000 q draws · seeds402/403 · separate wide-domain numerical integration

A random draw is not automatically a differentiable path



> Which connection would the derivative need to follow?

A random draw is not automatically a differentiable path




```
u = Normal(mu, sigma).sample()
u.requires_grad   # False
```

Does this sample retain that path in PyTorch?

Ask the sample itself for a gradient

```
u = Normal(mu, sigma).sample()
u.backward()
```

What do you predict before running backward?

Ask the sample itself for a gradient

```
u = Normal(mu, sigma).sample()
u.backward()
```

RuntimeError: this tensor does not require grad.

What has this failure shown?

A backward call can run and still miss the intended path

```
u = q.sample((S,))
loss = (q.log_prob(u) - p.log_prob(u)).mean()
loss.backward()
```

Can log q(u) depend on μ and σ even if u is detached?

A backward call can run and still miss the intended path

```
u = q.sample((S,))
loss = (q.log_prob(u) - p.log_prob(u)).mean()
loss.backward()
```

log q still contains the parameters explicitly. The missing sample path has not been repaired.

> Would a nonempty gradient tensor prove this is the desired KL derivative?

Check the derivative, not just the existence of a tensor

Derivative route	∂L/∂μ	∂L/∂σ
Naive detached samples	0.0013	−0.0003
Pathwise samples	−0.0622	−0.3752
Analytic KL	−0.0625	−0.3750

> Which row agrees with the derivative of the actual KL?

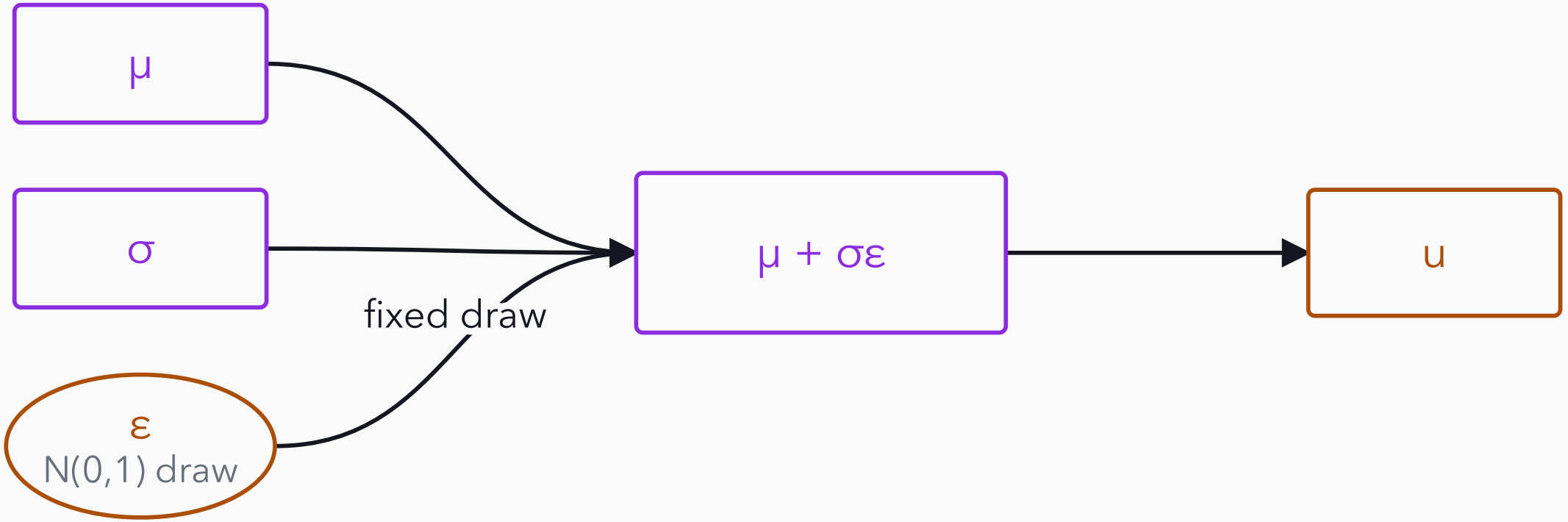
Check the derivative, not just the existence of a tensor

Derivative route	∂L/∂μ	∂L/∂σ
Naive detached samples	0.0013	−0.0003
Pathwise samples	−0.0622	−0.3752
Analytic KL	−0.0625	−0.3750

Same 50,000 chosen noise values · seed13 · q=Normal(0,2), p=Normal(1,4)

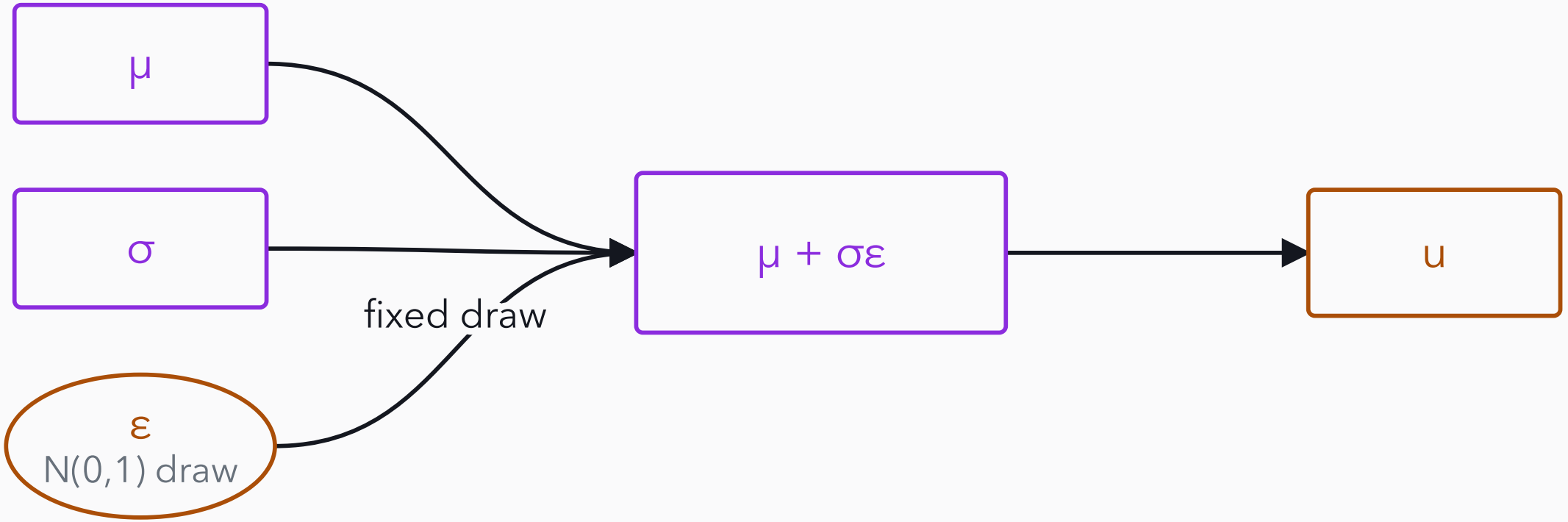
> Is Monte Carlo agreement expected to be exact?

Keep the noise separate from the parameters



Which random distribution is independent of μ and σ?

Keep the noise separate from the parameters



$$\epsilon \sim \mathcal{N}(0,1), \qquad u = \mu + \sigma\epsilon$$

For a fixed ε, which part is now deterministic?

What happens to this same draw when a parameter changes?

$$\epsilon = -0.75, \quad \mu = 0, \quad \sigma = 2 \quad \Longrightarrow \quad u = -1.5$$

> Hold ε fixed. What is u?

Chosen scalar input ε=−0.75 · arithmetic and autograd checked

What happens to this same draw when a parameter changes?

$$\epsilon = -0.75, \quad \mu = 0, \quad \sigma = 2 \quad \Longrightarrow \quad u = -1.5$$

$$\frac{\partial u}{\partial \mu} = 1$$

> If μ increases by a small amount, how does u change?

Chosen scalar input ε=−0.75 · arithmetic and autograd checked

What happens to this same draw when a parameter changes?

$$\epsilon = -0.75, \quad \mu = 0, \quad \sigma = 2 \quad \Longrightarrow \quad u = -1.5$$

$$\frac{\partial u}{\partial \mu} = 1$$

$$\frac{\partial u}{\partial \sigma} = \epsilon = -0.75$$

> Why can increasing σ move this particular u left?

Chosen scalar input ε=−0.75 · arithmetic and autograd checked

The loss gradient follows the same draw

$$L = (u - 1)^2, \qquad \frac{\partial L}{\partial u} = 2(u - 1) = -5$$

At u=−1.5, what derivative reaches the sampled value?

The loss gradient follows the same draw

$$L = (u-1)^2, \qquad \frac{\partial L}{\partial u} = 2(u-1) = -5$$

$$\frac{\partial L}{\partial \mu} = -5, \qquad \frac{\partial L}{\partial \sigma} = (-5)(-0.75) = 3.75$$

How does that derivative reach μ and σ?

A fixed-noise finite difference checks the path

```
q = Normal(mu, sigma)
u = q.rsample((S,))
loss = (q.log_prob(u) - p.log_prob(u)).mean()
```

What is different about rsample?

A fixed-noise finite difference checks the path

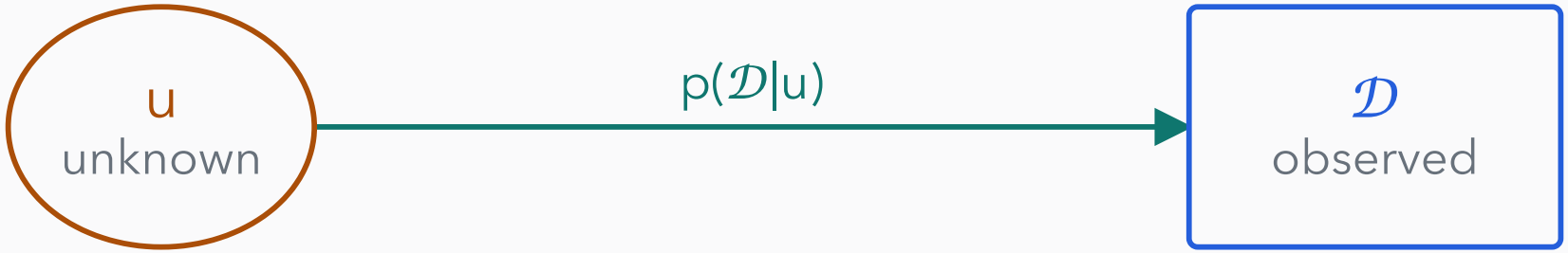
```
q = Normal(mu, sigma)
u = q.rsample((S,))
loss = (q.log_prob(u) - p.log_prob(u)).mean()
```

Derivative	Autograd	Finite difference
μ	−0.062466	−0.062466
σ	−0.372223	−0.372223

What must stay fixed on both sides of the finite difference?

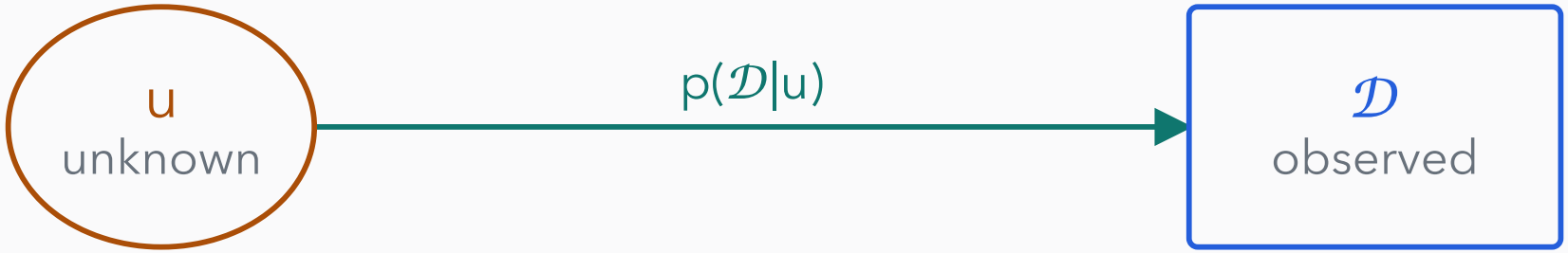
Measured check · seed88 · S=2048 · finite-difference $h=10^{-5}$

Now the target is an unknown posterior



> The model generates data from u. What is the backwards question?

Now the target is an unknown posterior



Inference asks for $p(u \mid \mathcal{D})$

Which object is our new target distribution?

Choose a family we can draw from and evaluate

$$q_\psi(u)$$

What is ψ? What is u?

Choose a family we can draw from and evaluate

$$q_\psi(u)$$

$$u \sim q_\psi, \qquad \log q_\psi(u)$$

> What operations do we need from this chosen family?

The direct score asks for the density we do not have

$$\mathrm{KL}(q_\psi(u)\|p(u \mid \mathcal{D})) = \mathbb{E}_{u\sim q_\psi}[\log q_\psi(u) - \log p(u \mid \mathcal{D})]$$

> Which pieces can we already compute?

The direct score asks for the density we do not have

$$\mathrm{KL}(q_\psi(u)\|p(u \mid \mathcal{D})) = \mathbb{E}_{u\sim q_\psi}[\log q_\psi(u) - \log p(u \mid \mathcal{D})]$$

The unavailable term is $\log p(u \mid \mathcal{D})$.

> Which piece blocks direct evaluation of this score?

Replace the posterior with joint divided by evidence

$$p(u \mid \mathcal{D}) = \frac{p(\mathcal{D}, u)}{p(\mathcal{D})}$$

> Which factor can be built from prior and likelihood?

Replace the posterior with joint divided by evidence

$$p(u \mid \mathcal{D}) = \frac{p(\mathcal{D}, u)}{p(\mathcal{D})}$$

$$p(\mathcal{D}, u) = p(\mathcal{D} \mid u)\, p(u)$$

> Which denominator may still be difficult?

The log quotient becomes a difference

$$\log p(u \mid \mathcal{D}) = \log p(\mathcal{D}, u) - \log p(\mathcal{D})$$

> What is the log of joint divided by evidence?

The log quotient becomes a difference

$$\log p(u \mid \mathcal{D}) = \log p(\mathcal{D}, u) - \log p(\mathcal{D})$$

$$-\log p(u \mid \mathcal{D}) = -\log p(\mathcal{D}, u) + \log p(\mathcal{D})$$

Why does evidence become positive when the posterior term is subtracted?

Keep the KL; change only the posterior term

$$\mathrm{KL}(q_\psi(u)\|p(u \mid \mathcal{D})) = \mathbb{E}_{u\sim q_\psi}[\log q_\psi(u) - \log p(u \mid \mathcal{D})]$$

Which expression stays the same on the left?

Keep the KL; change only the posterior term

$$\mathrm{KL}(q_\psi(u)\|p(u \mid \mathcal{D})) = \mathbb{E}_{u\sim q_\psi}[\log q_\psi(u) - \log p(u \mid \mathcal{D})]$$

$$= \mathbb{E}_{u\sim q_\psi}[\log q_\psi(u) - \log p(\mathcal{D}, u) + \log p(\mathcal{D})]$$

Which three terms now appear inside the average?

Constant with respect to what?

$$\mathbb{E}_{u\sim q_\psi}\left[\log q_\psi(u) - \log p(\mathcal{D}, u) + \log p(\mathcal{D})\right]$$

> What varies inside this expectation?

Constant with respect to what?

$$\mathbb{E}_{u\sim q_\psi}\left[\log q_\psi(u) - \log p(\mathcal{D}, u) + \log p(\mathcal{D})\right]$$

The model and observed data are fixed. Evidence depends on neither u nor ψ.

> Does log evidence vary with the sampled u or the fitted ψ?

Constant with respect to what?

$$\mathbb{E}_{u\sim q_\psi}\left[\log q_\psi(u) - \log p(\mathcal{D}, u) + \log p(\mathcal{D})\right]$$

The model and observed data are fixed. Evidence depends on neither u nor ψ.

$$= \mathbb{E}_{u\sim q_\psi}\left[\log q_\psi(u) - \log p(\mathcal{D}, u)\right] + \log p(\mathcal{D})$$

> What justifies moving it outside?

Swap sides and reverse the bracket's signs

$$\mathrm{KL}(q_\psi(u)\|p(u \mid \mathcal{D})) = \mathbb{E}_{u\sim q_\psi}[\log q_\psi(u) - \log p(\mathcal{D}, u)] + \log p(\mathcal{D})$$

After rearranging, what is the sign of the joint term?

Swap sides and reverse the bracket's signs

$$\mathrm{KL}(q_\psi(u)\|p(u \mid \mathcal{D})) = \mathbb{E}_{u\sim q_\psi}[\log q_\psi(u) - \log p(\mathcal{D}, u)] + \log p(\mathcal{D})$$

$$\log p(\mathcal{D}) = \mathbb{E}_{u\sim q_\psi}[\log p(\mathcal{D}, u) - \log q_\psi(u)] + \mathrm{KL}(q_\psi(u)\|p(u \mid \mathcal{D}))$$

Which term remains as a nonnegative gap?

A trainable objective emerges from the rearrangement

$$\mathrm{ELBO}(q_\psi) = \mathbb{E}_{u \sim q_\psi}\left[\log p(\mathcal{D}, u) - \log q_\psi(u)\right]$$

> Which density has disappeared from the computable expression?

A trainable objective emerges from the rearrangement

$$\mathrm{ELBO}(q_\psi) = \mathbb{E}_{u \sim q_\psi}\left[\log p(\mathcal{D}, u) - \log q_\psi(u)\right]$$

Evaluate log joint and log q at q's draws. Average to estimate the expectation.

> What do we still need to estimate in a general problem?

One fixed total, two parts

$$\log p(\mathcal{D}) = \mathrm{ELBO}(q_\psi) + \underbrace{\mathrm{KL}(q_\psi(u) \| p(u \mid \mathcal{D}))}_{\geq 0}$$

Why can the ELBO not exceed the log evidence?

One fixed total, two parts

$$\log p(\mathcal{D}) = \mathrm{ELBO}(q_\psi) + \underbrace{\mathrm{KL}(q_\psi(u) \| p(u \mid \mathcal{D}))}_{\geq 0}$$

$$\mathrm{ELBO}(q_\psi) \leq \log p(\mathcal{D})$$

When would equality hold?

One fixed total, two parts

$$\log p(\mathcal{D}) = \mathrm{ELBO}(q_\psi) + \underbrace{\mathrm{KL}(q_\psi(u) \| p(u \mid \mathcal{D}))}_{\geq 0}$$

$$\mathrm{ELBO}(q_\psi) \leq \log p(\mathcal{D})$$

Raise the ELBO; lower the posterior KL by the same amount.

> Which optimisation direction improves q?

Optimise the ELBO without computing the evidence

$$\max_{\psi} \mathrm{ELBO}(q_\psi) \quad \Longleftrightarrow \quad \min_{\psi} [-\mathrm{ELBO}(q_\psi)]$$

> Is the unknown evidence being set to zero?

Optimise the ELBO without computing the evidence

$$\max_{\psi} \mathrm{ELBO}(q_\psi) \quad \Longleftrightarrow \quad \min_{\psi} [-\mathrm{ELBO}(q_\psi)]$$

Need: draws from q, log q, log joint.

Do not need: the normalised posterior or its evidence.

> What exactly makes it safe to omit from ψ's optimisation?

Return to the same coin and the same data

H H T H T H T H H H T T H T H

H T H T T H H H T T H T H T H

$$p(r) = \mathrm{Beta}(5, 3)$$

What are we holding fixed from the opening problem?

Return to the same coin and the same data

H H T H T H T H H H T T H T H

H T H T T H H H T T H T H T H

$$p(r) = \mathrm{Beta}(5, 3)$$

Fit q using the joint. Keep the known posterior for validation.

> Why retain a known answer if the optimiser does not use it?

The conjugate posterior is a reference answer

$$p(r \mid \mathcal{D}) = \mathrm{Beta}(5 + 17,\ 3 + 13)$$

> Which prior parameter receives the heads count?

The conjugate posterior is a reference answer

$$p(r \mid \mathcal{D}) = \mathrm{Beta}(5 + 17,\ 3 + 13)$$

$$= \mathrm{Beta}(22, 16)$$

> Does the optimiser need to call this posterior density?

Keep the prior’s normalising constant

$$p(r) = \frac{r^4(1-r)^2}{B(5,3)}$$

> Why are the exponents 4 and 2 rather than 5 and 3?

Keep the prior’s normalising constant

$$p(r) = \frac{r^4(1-r)^2}{B(5,3)}$$

$$\log p(r) = 4\log r + 2\log(1-r) - \log B(5,3)$$

> Can we drop this constant when checking absolute ELBO values?

Each observed toss contributes one log probability

$$\log p(\mathcal{D} \mid r) = 17 \log r + 13 \log(1 - r)$$

What contributes log r? What contributes log(1−r)?

Each observed toss contributes one log probability

$$\log p(\mathcal{D} \mid r) = 17 \log r + 13 \log(1 - r)$$

One ordered sequence. There is no binomial coefficient in this convention.

> Why is there no factor counting different orders?

The joint is available at any proposed r

$$\log p(\mathcal{D}, r) = \log p(\mathcal{D} \mid r) + \log p(r)$$

> Which two available functions appear here?

The joint is available at any proposed r

$$\log p(\mathcal{D}, r) = \log p(\mathcal{D} \mid r) + \log p(r)$$

```
log_likelihood = 17*log(r) + 13*log1p(-r)
log_joint = prior.log_prob(r) + log_likelihood
```

Where is the unknown evidence in this code?

Fit positive Beta shapes through unconstrained parameters

$$q_\psi(r) = \mathrm{Beta}(a, b)$$

> What family contains our exact posterior in this toy?

Fit positive Beta shapes through unconstrained parameters

$$q_\psi(r) = \mathrm{Beta}(a, b)$$

$$a = \exp(\psi_a), \qquad b = \exp(\psi_b)$$

> Which quantities does the optimiser store and update?

One mean over draws; the data are already summed

```
a, b = psi.exp()
q = Beta(a, b)
r = q.rsample((512,))
terms = log_joint(r) - q.log_prob(r)
loss = -terms.mean()
```

Where are the thirty observation contributions combined?

One mean over draws; the data are already summed

```
a, b = psi.exp()
q = Beta(a, b)
r = q.rsample((512,))
terms = log_joint(r) - q.log_prob(r)
loss = -terms.mean()
```

r and terms each have shape [512]. Mean over Monte Carlo draws.

> Why is there no additional sum over the 512 draws before mean?

Inspect one actual update

Quantity	Before	After one Adam step
a	10.0000	10.2532
b	10.0000	9.7531

> Which shape parameter should initially increase for more heads?

Inspect one actual update

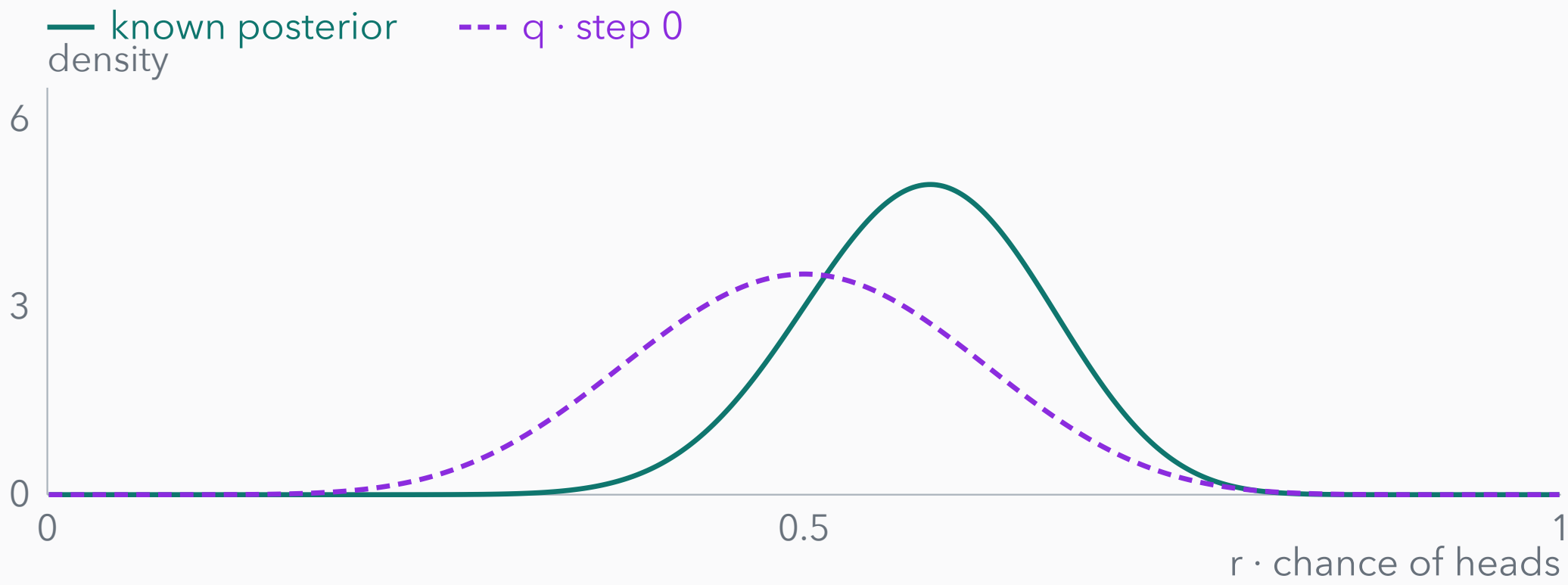
Quantity	Before	After one Adam step
a	10.0000	10.2532
b	10.0000	9.7531

$$\nabla_{\psi}[-\widehat{\mathrm{ELBO}}] \approx (-3.338,\ 2.960)$$

> Are the printed gradients with respect to a,b or ψ?

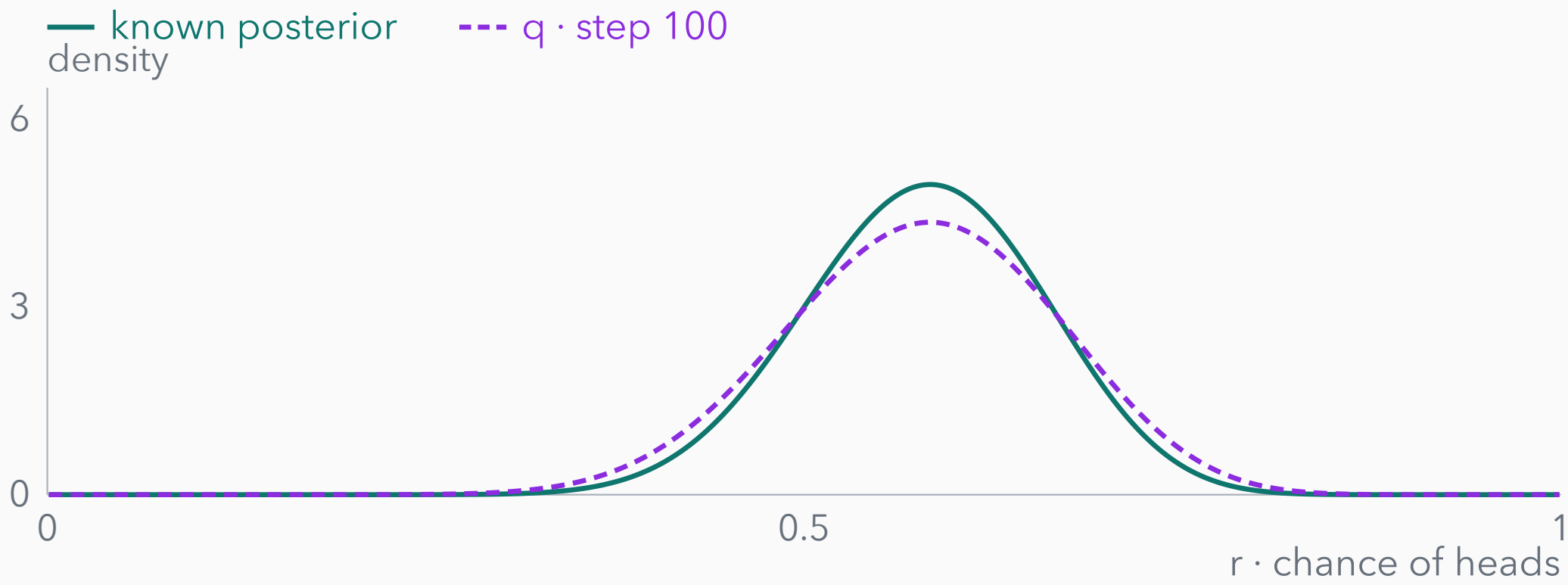
Measured first update · seed2401 · Adam lr=0.025 · 512 Beta draws

A measured fit using only the joint



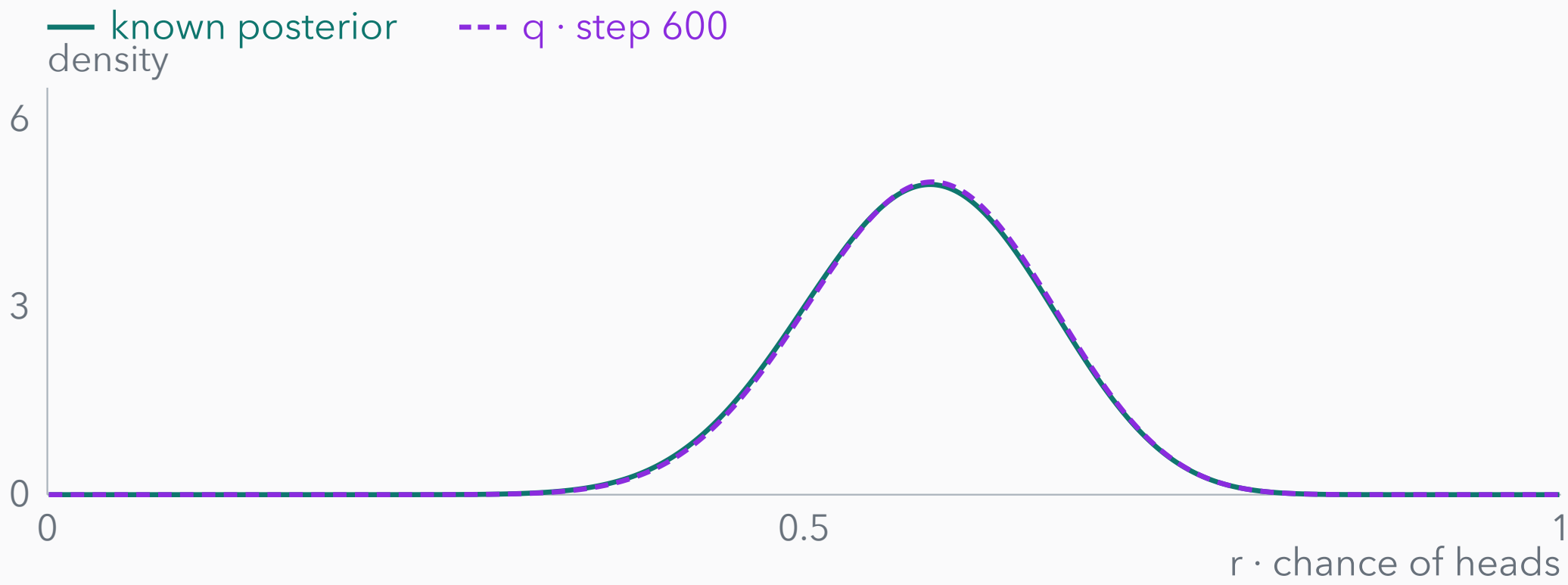
Where does the initial q put its mass?

A measured fit using only the joint



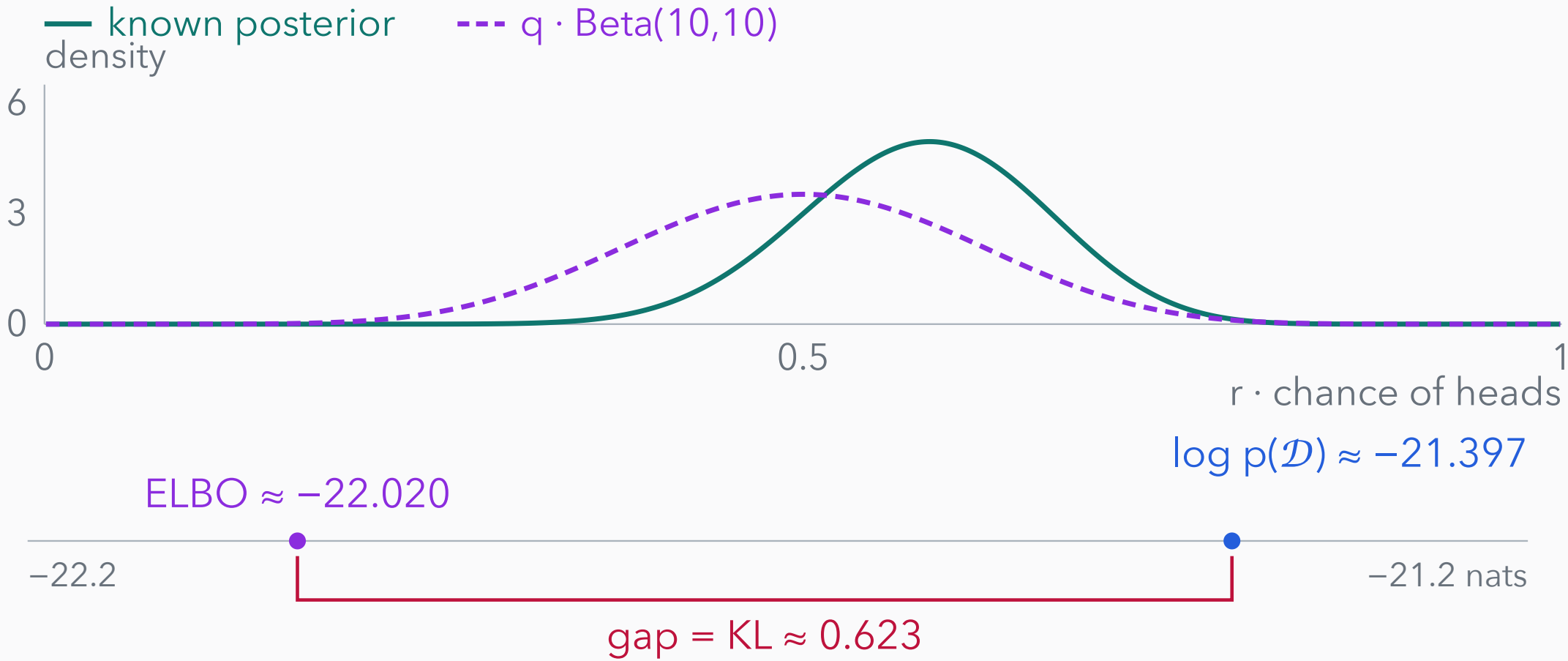
What changed by step100?

A measured fit using only the joint



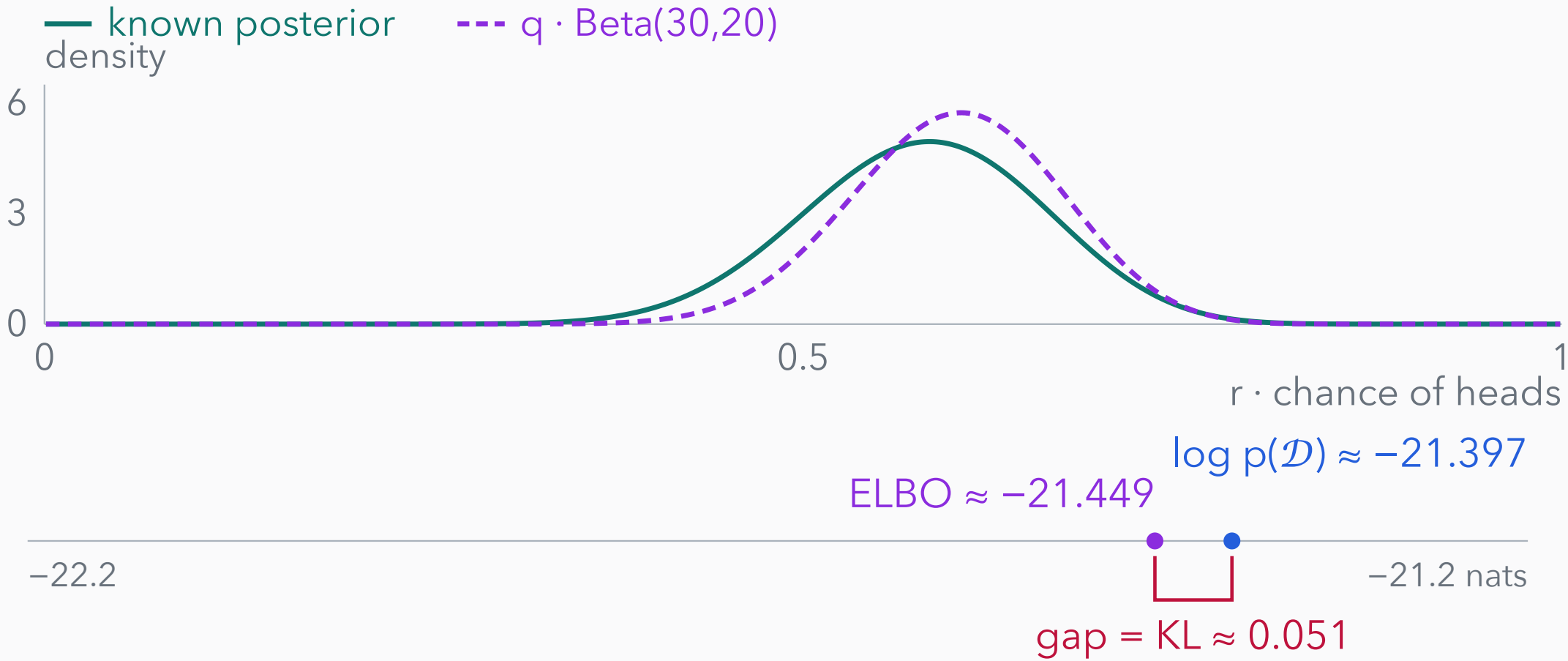
Is the final q exactly the known posterior?

Check the fixed total with three selected candidates



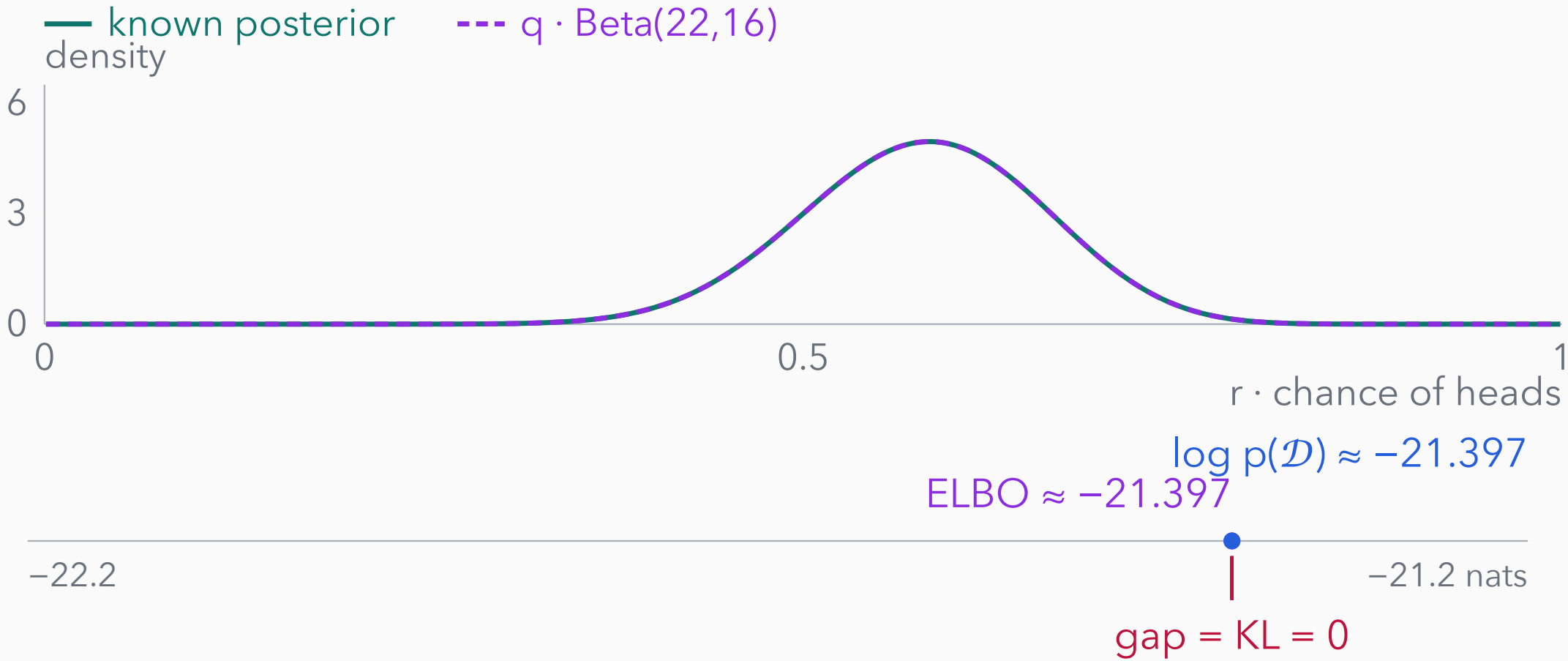
Add the ELBO and the gap. What is fixed?

Check the fixed total with three selected candidates



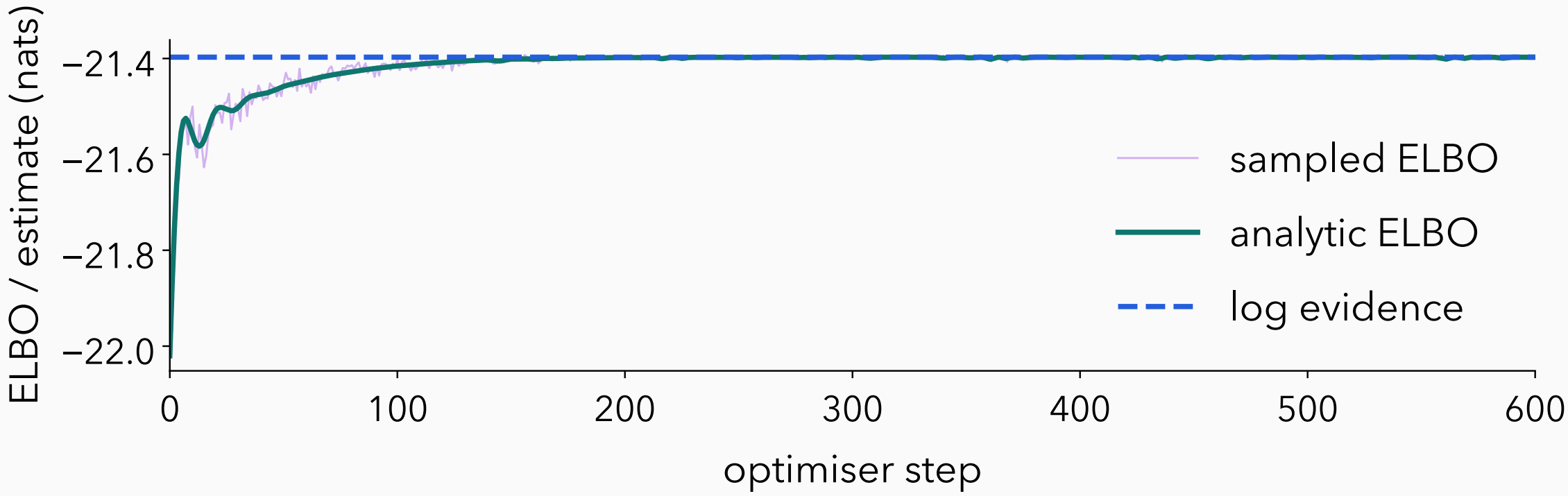
Does the better candidate raise the ELBO by the amount its KL falls?

Check the fixed total with three selected candidates



What happens at the exact posterior?

An estimate is not itself guaranteed to be a lower bound

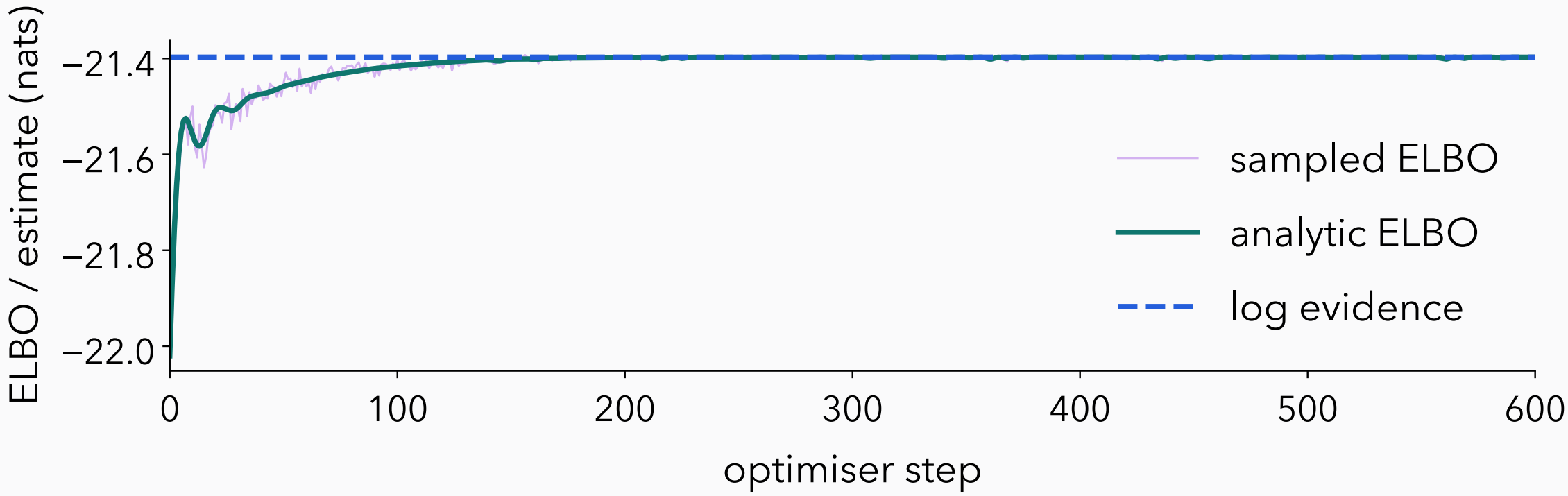


Same pre-update q at each step · fresh training draws · known log evidence reference

> Can a finite-sample ELBO estimate cross the reference line?

Measured seed2401 training trace · sampled objective versus direct analytic diagnostic

An estimate is not itself guaranteed to be a lower bound



Same pre-update q at each step · fresh training draws · known log evidence reference

162 of 600 sampled estimates lie above the known evidence in this run.

> Which object remains a mathematical lower bound?

A higher or stable ELBO does not prove a zero gap

$$\mathrm{ELBO}(q_{\mathrm{new}}) > \mathrm{ELBO}(q_{\mathrm{old}})$$

> What does a true increase in the exact ELBO establish?

A higher or stable ELBO does not prove a zero gap

$$\mathrm{ELBO}(q_{\mathrm{new}}) > \mathrm{ELBO}(q_{\mathrm{old}})$$

The family may miss the posterior. Optimisation may stop short.

> Does a stable loss establish q=p(u|$\mathcal{D}$)?

For KL(q||p), where do the samples come from?

$$\mathrm{KL}(q\|p) = \boxed{\quad ? \quad}[\log q(u) - \log p(u)]$$

> Answer without looking at the earlier definition.

For KL(q||p), where do the samples come from?

$$\mathrm{KL}(q\|p) = \boxed{\quad ? \quad}[\log q(u) - \log p(u)]$$

$$\boxed{\mathbb{E}_{u\sim q}}$$

Why does sampling from p change the question?

What changes during fixed-model variational inference?

Object	Changes during this inference fit?
Observed data $\mathcal{D}$	?
Exact posterior p(u|$\mathcal{D}$)	?
Variational parameters ψ	?
A newly drawn u	?

> Which objects change, and in what sense?

What changes during fixed-model variational inference?

Object	Changes during this inference fit?
Observed data $\mathcal{D}$	?
Exact posterior $p(u\|\mathcal{D})$	?
Variational parameters ψ	?
A newly drawn u	?

Update ψ. Draw fresh u. The fixed data, model and exact posterior do not change.

> Is a new draw an optimiser parameter update?

Why can we omit the evidence from ψ's optimisation?

$$\mathrm{KL}(q_\psi \| p(u \mid \mathcal{D})) = -\mathrm{ELBO}(q_\psi) + \log p(\mathcal{D})$$

> Is the reason that evidence is zero, easy, or constant?

Why can we omit the evidence from ψ's optimisation?

$$\mathrm{KL}(q_\psi \| p(u \mid \mathcal{D})) = -\mathrm{ELBO}(q_\psi) + \log p(\mathcal{D})$$

$$\nabla_\psi \log p(\mathcal{D}) = 0$$

> Constant with respect to which parameters?

A sensor has an unknown constant bias

$$u \sim \mathcal{N}(0, 1), \qquad y_i \mid u \sim \mathcal{N}(u, 0.5^2)$$

Identify the unknown, observations and one possible q family.

A sensor has an unknown constant bias

$$u \sim \mathcal{N}(0, 1), \qquad y_i \mid u \sim \mathcal{N}(u, 0.5^2)$$

$$p(\mathcal{D}, u) = p(u) \prod_i p(y_i \mid u)$$

> Which available functions form the joint?

A sensor has an unknown constant bias

$$u \sim \mathcal{N}(0, 1), \qquad y_i \mid u \sim \mathcal{N}(u, 0.5^2)$$

$$p(\mathcal{D}, u) = p(u) \prod_i p(y_i \mid u)$$

$$\max_{\psi} \mathbb{E}_{q_\psi(u)}[\log p(\mathcal{D}, u) - \log q_\psi(u)]$$

What objective would you implement?

Generative models will create this inference problem again

A hidden description can generate an observation.

> Where might a new unknown appear in an image model?

Generative models will create this inference problem again

A hidden description can generate an observation.

Next: learn representations for images, then ask whether they can generate new images.

> What question should we ask about a good reconstruction?

Where does the complete Gaussian KL formula come from?

$$\log \frac{q(u)}{p(u)} = \log \frac{s}{\sigma} + \frac{(u-m)^2}{2s^2} - \frac{(u-\mu)^2}{2\sigma^2}$$

> Which log-normalising constants cancel?

Where does the complete Gaussian KL formula come from?

$$\log \frac{q(u)}{p(u)} = \log \frac{s}{\sigma} + \frac{(u-m)^2}{2s^2} - \frac{(u-\mu)^2}{2\sigma^2}$$

$$\mathbb{E}_q[(u-m)^2] = \sigma^2 + (\mu - m)^2, \quad \mathbb{E}_q[(u-\mu)^2] = \sigma^2$$

> Which two moments are needed?

Where does the complete Gaussian KL formula come from?

$$\log \frac{q(u)}{p(u)} = \log \frac{s}{\sigma} + \frac{(u-m)^2}{2s^2} - \frac{(u-\mu)^2}{2\sigma^2}$$

$$\mathbb{E}_q[(u-m)^2] = \sigma^2 + (\mu - m)^2, \quad \mathbb{E}_q[(u-\mu)^2] = \sigma^2$$

$$\mathrm{KL}(q\|p) = \log \frac{s}{\sigma} + \frac{\sigma^2 + (\mu-m)^2}{2s^2} - \frac{1}{2}$$

Where does the final −1/2 come from?

Why is exact KL nonnegative?

$$\mathrm{KL}(q\|p) = -\mathbb{E}_q\left[\log \frac{p(u)}{q(u)}\right]$$

Which function makes Jensen's inequality useful?

Why is exact KL nonnegative?

$$\mathrm{KL}(q\|p) = -\mathbb{E}_q\left[\log \frac{p(u)}{q(u)}\right]$$

$$\geq -\log \mathbb{E}_q\left[\frac{p(u)}{q(u)}\right]$$

Why does the inequality point this way?

Why is exact KL nonnegative?

$$\mathrm{KL}(q\|p) = -\mathbb{E}_q\left[\log \frac{p(u)}{q(u)}\right]$$

$$\geq -\log \mathbb{E}_q\left[\frac{p(u)}{q(u)}\right]$$

$$= -\log 1 = 0$$

What makes the final expectation one?

Pathwise gradients are one route, not the only route

$$\nabla_\psi \mathbb{E}_{q_\psi}[f_\psi(u)] = \mathbb{E}_{q_\psi}[f_\psi(u)\nabla_\psi \log q_\psi(u) + \nabla_\psi f_\psi(u)]$$

> Which term was missing from naive detached-sample differentiation?

Differentiation-under-the-integral identity · regularity assumptions apply

Pathwise gradients are one route, not the only route

$$\nabla_\psi \mathbb{E}_{q_\psi}[f_\psi(u)] = \mathbb{E}_{q_\psi}[f_\psi(u)\nabla_\psi \log q_\psi(u) + \nabla_\psi f_\psi(u)]$$

A score-function estimator accounts explicitly for the changing sampling distribution.

> Does this identity require u=μ+σε?

Differentiation-under-the-integral identity · regularity assumptions apply

A mean must say which dimension it averages

Object	Shape	Correct reduction
S scalar Beta draws	[S]	mean over S
S d-dimensional joint log densities	[S]	mean over S
S factorised coordinate log densities	[S,d]	sum event d, then mean S

> Does every log_prob output still have an event dimension?

A mean must say which dimension it averages

Object	Shape	Correct reduction
S scalar Beta draws	[S]	mean over S
S d-dimensional joint log densities	[S]	mean over S
S factorised coordinate log densities	[S,d]	sum event d, then mean S

MultivariateNormal.log_prob already sums the event. Do not sum the sample axis a second time.

> What goes wrong if you sum S before subtracting log q from the joint?

An ordered sequence and a count have different evidence

$$p(H = 17 \mid r) = \binom{30}{17} r^{17}(1-r)^{13}$$

> Which event contains more possible sequences?

An ordered sequence and a count have different evidence

$$p(H = 17 \mid r) = \binom{30}{17} r^{17}(1-r)^{13}$$

Observed object	Log evidence
One ordered sequence	−21.39724
Only the head count	−2.79624

> Does this constant change the posterior or its KL gap?

Exact convention comparison · log binomial coefficient computed with log-gamma

Check the sampled objective with analytic Beta moments

$$\mathbb{E}_q[\log r] = \operatorname{digamma}(a) - \operatorname{digamma}(a+b)$$

> Which two expectations determine the coin ELBO?

Check the sampled objective with analytic Beta moments

$$\mathbb{E}_q[\log r] = \mathrm{digamma}(a) - \mathrm{digamma}(a+b)$$

$$\mathbb{E}_q[\log(1-r)] = \mathrm{digamma}(b) - \mathrm{digamma}(a+b)$$

> Why write digamma as a name rather than ψ here?

Check the sampled objective with analytic Beta moments

$$\mathbb{E}_q[\log r] = \mathrm{digamma}(a) - \mathrm{digamma}(a+b)$$

$$\mathbb{E}_q[\log(1-r)] = \mathrm{digamma}(b) - \mathrm{digamma}(a+b)$$

Insert these into expected log likelihood + expected log prior − expected log q.

> How was the identity checked non-circularly?